Diseases of small cardamom

1. Rhizome rot /root rot/damping off clump rot - *Pythium vexans, Rhizoctonia solani Fusarium oxysporum*
2. Azhukal disease or capsule rot - ***Phytophthora nicotianae* var. *nicotianae***, *Pythium vexans, P. meadii* and *Fusarium* spp
3. Katte or marble or mosaic disease - *Cardamom Mosaic Virus* (*CdMV*)

Diseases of large cardamom

1. Chirkey disease - *Large cardamom chirke virus (LCCV)*
2. Foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

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3. Mosaic or katte disease - *Cardamom Mosaic Virus* (*CdMV*)

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- Small cardamom (*Elettaria cardamomum* Maton) is cultivated mainly in the **forests of Western Ghats of the country**
- **Rhizome rot is one of the oldest endemic fungal disease** occurring in different parts of Kerala and Karnataka and is also called **clump rot.**
- The disease has also become increasingly important in **nurseries causing damping-off.**
- The disease incidence ranges from **9 to 60 per cent** in different areas of Idukki.

Small cardamom rhizome rot /root rot/Clump rot/damping off - *Pythium vexans, Rhizoctonia solani, Fusarium oxysporum*

Symptoms:

- The foliage of the infected plants becomes **yellow followed by drooping of leaves.**
- The infection is initiated **at the collar region which breaks off even with slight disturbance.**
- As the disease advances, **rotting of the collar region starts** which extends to the rhizomes and roots and the affected tillers later fall off.
- **Rhizome rot and shoot lodging are severe during monsoon season**.
- In seedlings, the **damping off symptoms are produced and these die and collapse in patches.**

Small cardamom rhizome rot /root rot/Clump rot/damping off - *Pythium vexans, Rhizoctonia solani, Fusarium oxysporum*

Pathogen:

- **Rhizome rot** - *Pythium vexans, Rhizoctonia solani*
- **Seed and seedling rot** - *Fusarium oxysporum*

Disease cycle and epidemiology:

- Pathogens responsible for this disease are soil borne and the apparently healthy seedlings from infected nurseries serve as the carrier of the inoculum to the main field.
- **Poor drainage** mainly predisposes the disease initiation.
- Intensity of rainfall influences the disease development and **more than 80 per cent infection occurs in between June-October.**
- **Nematode (*Meloidogyne incognita*) predisposes the seedlings to damping off or rhizome rot caused by *Rhizoctonia solani*.**

Small cardamom rhizome rot /root rot/Clump rot/damping off - *Pythium vexans, Rhizoctonia solani, Fusarium oxysporum*

Management

- **Rogueing of infected plants** in older plantations and change the site of nursery.
- **Proper drainage**.
- Apply **pongamia cake (250g/m2) and Eupatorium leaves** as mulch is useful in reducing the damping-off.
- **Treat nursery bed soil with Formaline (5%) three weeks before transplanting.**
- Treat seed with carbendazim +thiram (0.25%) or carbendazim + captan (0.25%) or captan alone (0.4%)
- With the initiation of the disease, spray the crop with the metalaxyl + mancozeb (0.25%) or Bordeaux mixture (4:4:50) and repeat at 10-12 days interval.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

The disease appears after the onset of South-West monsoon in the form of water soaked lesions on tender leaves, stem and capsules, which later form dead areas surrounded by yellow halo.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Azhukal disease / capsule rot / fruit rot

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

- The cultivar **Malabar is more severely affected.**
- **Azhukal,** a Malayalam name which literally **means rotting**, is a serious disease of this crop and causes a crop loss upto 30 per cent.
- During monsoon period, the disease appears **on tender leaves and capsules** as water- soaked lesions, which are surrounded by **yellow halo and later on these lesions turn black and the tissues become dead.**
- In advanced stages, the **infection spreads to panicles and tillers**.
- In extreme cases, the **rhizomes also rotting** results in the death of the plant.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Symptoms on Leaves:

- Symptoms can appear on **matured leaves, tillers, panicles and capsules.**
- Large circular, irregular, water soaked spots with **black colour appear on leaves surrounded by yellow halo.**
- **The exposed portion of the unopened leaves may rot**.
- Grey patches of **irregular spots with brown margin are formed at the base of the leaf sheath.**
- As a result, **the leaves rot and shred along the veins.**
- In the advanced stages, the affected **leaves break at the base of the petiole and remain hanging**

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Symptoms on capsule

- On the immature capsules, the symptoms develop as **water soaked discoloured areas, which later turn brownish.**
- Upon decay, such **capsules emit a foul smell** and subsequently drop off.
- Mature **capsules when infected, become shrivelled upon drying.**
- **Pseudostems and roots** are also infected.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Pathogen (s):

- Initially *Phytophthora nicotianae* var. *nicotianae*
- Later on, the involvement of *Pythium vexans, Phytophthora meadii* McRae A2 mating type and *Fusarium* spp. have also been reported.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Disease cycle and epidemiology:

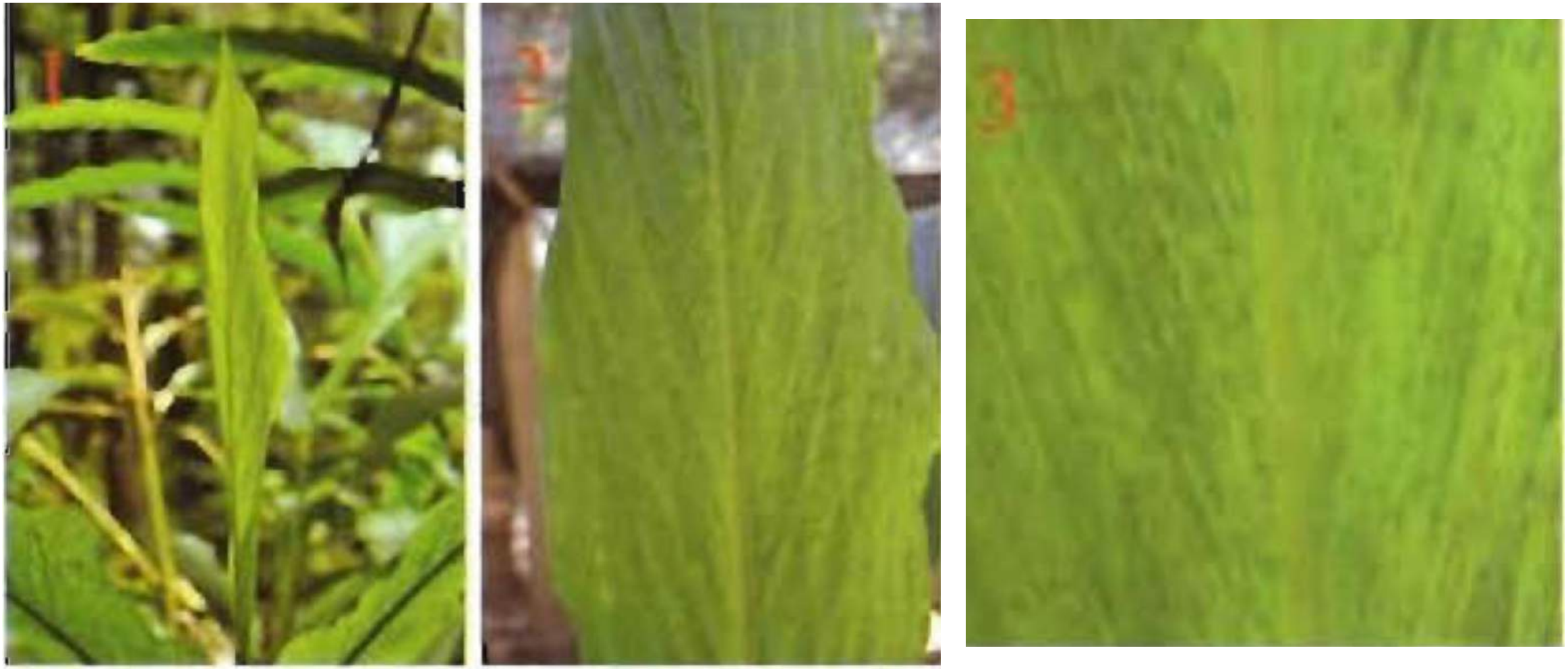
- The associated pathogen(s) are soil borne in nature and overwinter in the form of **oospores or chlamydospores.**
- **Colocasia plants are reported to be the collateral hosts of *P. meadii***
- Wet and humid weather conditions accompanied with low temperature prevailing in the plantations are highly favourable for the disease development.
- During August, the high rainfall and resultant increased relative humidity become the most congenial conditions for the disease development.

Small cardamom azhukal disease or capsule rot - *Phytophthora nicotianae* var. *nicotianae, P. meadii, Pythium vexans* and *Fusarium* spp

Management

- **Avoid movement of seed rhizomes from diseased area**
- Remove infected panicles and rhizomes.
- Improve drainage in the plantations.
- Soil amendment with neem cake has also been found effective (250 g/m^2).
- Biological control of *P. meadii* infections using antagonistic fungi like *Trichoderma* spp. and *Laetisaria* sp. has also been found effective.
- Application of fungicides like Bordeaux mixture (4:4:50) or copper oxychloride (0.3%) or metalaxyl + mancozeb (0.25%) or Aliette (0.3%) has been recommended.

Small cardamom marble or katte or mosaic disease - *CdMV*



Figs 1-3: Chlorotic flecks on young leaf;

Small cardamom marble or katte or mosaic disease - *CdMV*

Small cardamom marble or katte or mosaic disease - *CdMV*

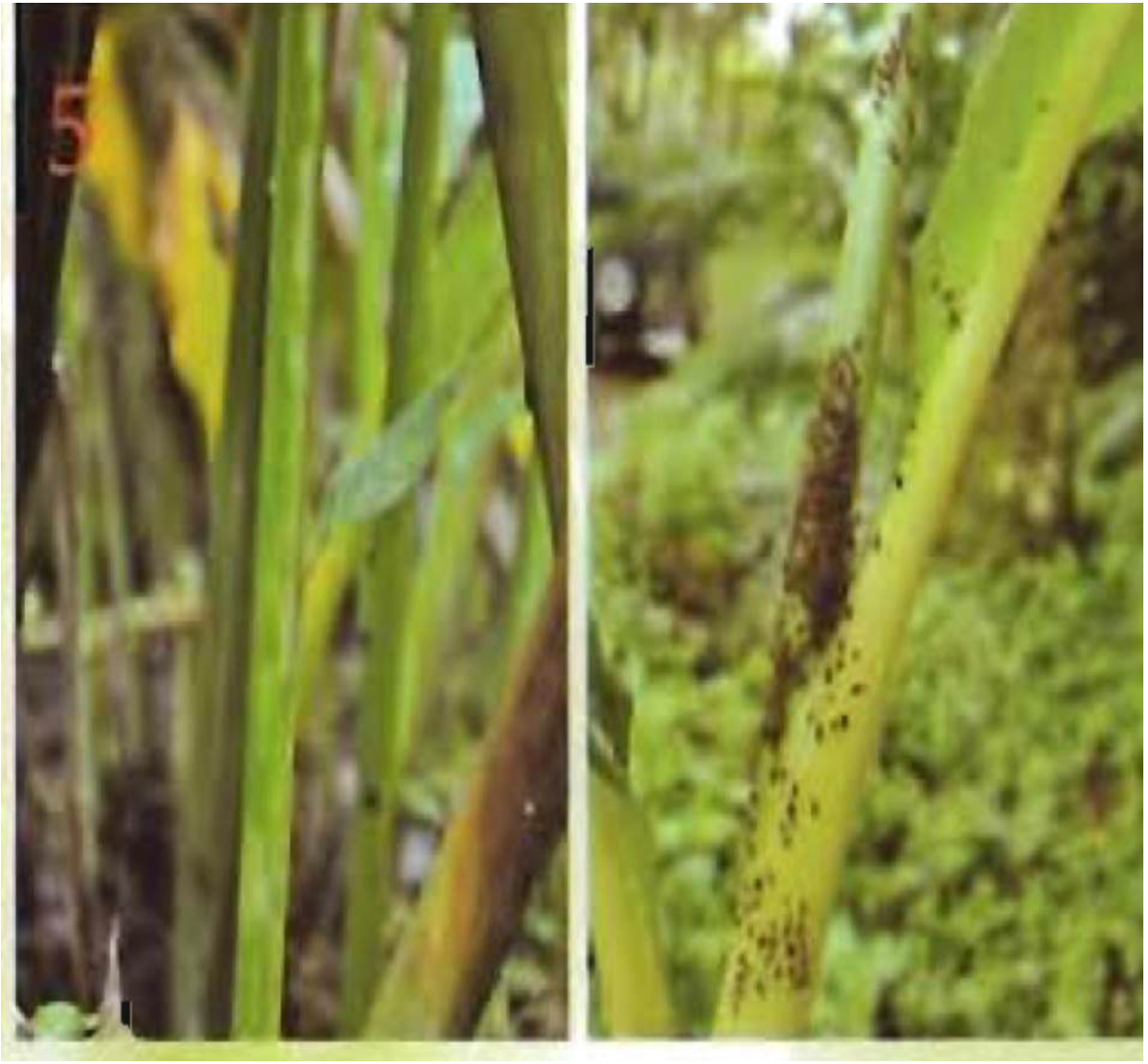


Fig 5: Mosaic mottling on leaf sheath and shoots; Fig 6: Aphid vector – Pentalonia caladii

Small cardamom marble or katte or mosaic disease - *CdMV*

- The disease is locally known as **Katte** meaning a disorder and is known as marble disease in **Anamalais.**
- The expression of symptoms vary with the varieties, growing region and strain
- The disease is widely distributed in cardamom growing tracts of India with incidence ranging from 0.01 to 99 per cent.
- The infection of plants at the seedling stage leads to cent per cent losses in yield.
- Symptoms in brief are severe mosaic, overall chlorosis, chlorotic flecks and necrotic and ring spots.

Small cardamom marble or katte or mosaic disease - *CdMV*

Symptoms:

- Slender chlorotic flecks measuring 2-5 mm in length appear on the youngest leaf of the affected tiller which later develop into discontinuous pale green stripes which run parallel to the veins from the midrib to leaf margin
- All new emerging leaves show characteristic mosaic symptoms with stripes of green tissue evenly distributed over the entire leaf lamina.
- The **mosaic symptoms are masked as the leaves mature.**
- **Mosaic type of mottling is often seen on the leaf sheaths and young pseudostems.**

Small cardamom marble or katte or mosaic disease - *CdMV*

Symptoms contd….

- The infection is systemic in nature and gradually spreads to all the tillers in a plant.
- In advanced stages, the affected plants produce shorter and slender tillers with few shorter panicles and degenerate gradually.
- The plants infected by **mosaic or Katte can survive for many years and act as the source of inoculum**

Small cardamom marble or katte or mosaic disease - *CdMV*

Pathogen:

- Genus Macluravirus family Potyviridae which have 650 x 10-12 nm homogenous flexuous particles and inclusion bodies.

Disease cycle and epidemiology:

- Transmitted by banana aphid *Pentalonia nigronervosa* .
- Clones from infected gardens, seedlings raised in the vicinity of infected plantations, voluntary seedlings collected from infected plants and few infected zingiberaceous hosts form the primary source of disease introduction to cardamom plantations.
- Entry of the virus also occurs in contiguous and nearby plantations upto 1000 m from the external inoculum source.

Small cardamom marble or katte or mosaic disease - *CdMV*

Management

- Use healthy planting material which reduces the spread of the disease.
- **Periodic surveillance of the plantations** should be done **at least twice a year to detect the virus infection** and infected plants may be removed immediately.

Diseases of large cardamom

1. Chirkey disease - *Large cardamom chirke virus*
2. Foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Large cardamom chirkey disease - *Large cardamom chirke virus* (*LCCV*)

Fig1. *Chirke disease of large cardamom*

Large cardamom chirkey disease - *Large cardamom chirke virus* (*LCCV*)

Large cardamom chirkey disease - *Large cardamom chirke virus* (*LCCV*)

Symptoms:

- This is a major disease of large cardamom.
- The disease spreads rapidly and the symptoms can identified with keen observartion
- In young leaves, discrete pale green to yellow longitudinal stripes running parallel to each other can be seen, but in mature leaves, the symptoms are masked.
- The flecks measures 2.5 mm, the characteristic mosaic symptoms are also seen.
- The affected plants produce flowers and spikes but without capsules.

Large cardamom chirkey disease - *Large cardamom chirke virus* (*LCCV*)

Pathogen : Macluravirus, Potyviridae

The disease is caused by a ***Large cardamom chirke virus*** (***LCCV***)

Disease cycle and epidemiology

- The corn aphid *Ropholosiphum maydis*, *R. padi and Sitobion avene*
  The disease is also spread by planting infected suckers

Management:

- Removal of volunteers
- Use healthy planting material
- Control aphid vectors.

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

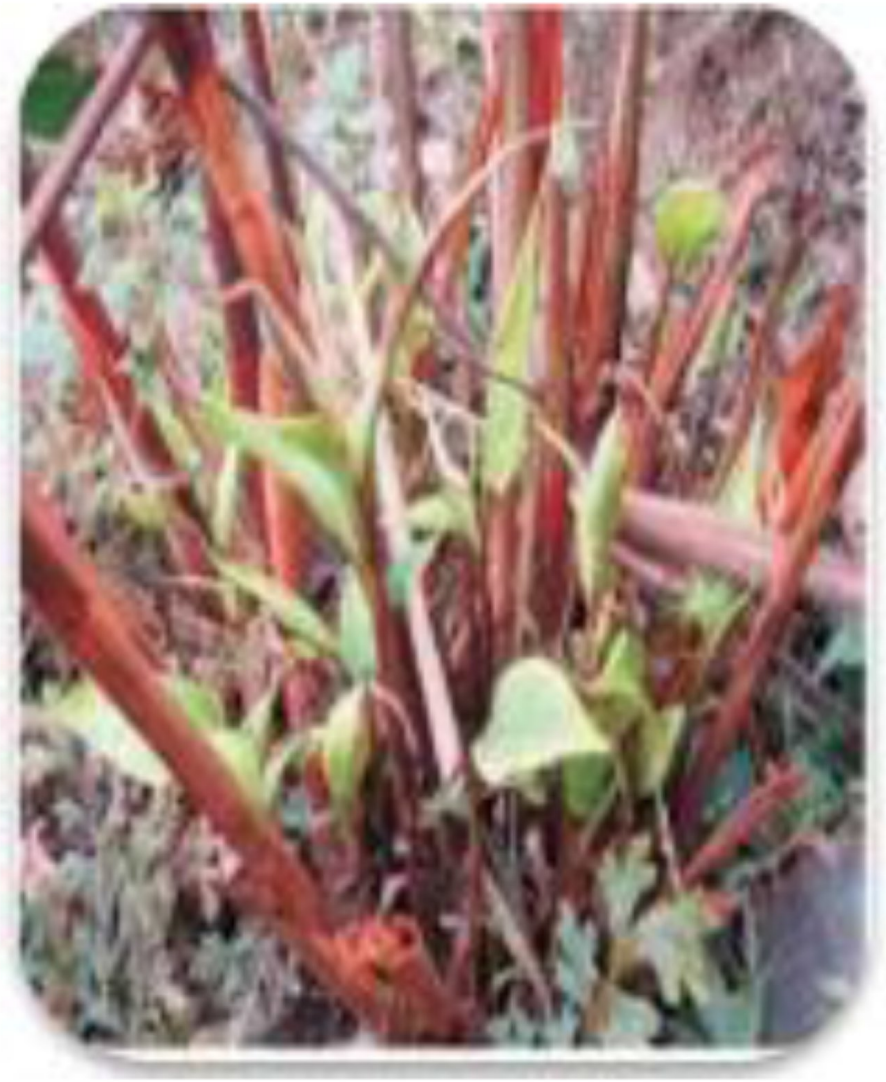


Fig2. *Foorkey disease of large cardamom*

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Symptoms

- Large cardamom (*Amomum subulatum*) is shade loving crop and is grown under humid and cool conditions.
- Sikkim has largest area under the crop in India
- The foorkey disease was first reported by Vasudeva in 1956.
- It causes severe damage in Darjeeling district of West Bengal

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Symptoms:

- Excessive sprouting and formation of bushy dwarf clumps at the base of mother plants that gradually die, characterize the disease.
- The leaves become small, lightly curled and pale green in colour.
- The inflorescence becomes stunted, thereby producing no flowers and fruits.
- The capsule size reduced and chaffy without seeds

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Pathogen : Nano virus Fy: Nanoviridae

- Disease is caused by *Cardamom bushy dwarf virus*
- Virus particles are 32 nm in diameter.

Disease cycle and epidemiology:

- Transmitted through banana aphid ***Mycromyzus kalimpongensis*** **and** ***Pentalonia nigronervosa***
- Diseased plants survive for few years but remain sterile and unproductive.

Large cardamom foorkey disease (dwarf bushy) - *Cardamom bushy dwarf virus*

Management:

- Infected plants should be uprooted and the surrounding are drenched with systemic insecticides to control movement of aphids on the main host as well as collateral hosts such as corn, banana, peach and squash.

- The plantation should be regularly sprayed once in three weeks with metasystox to check the movement of aphids.